



Effects of regional network economies on industrial productivity in Japan: dynamic total factor productivity function approach

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Abstract

Improving productivity performance is essential for achieving regional economic sustainability. With the advent of the network society, regional economic agents have benefited not only from local within-region interactions but also from external out-of-region economies. However, it remains unclear whether localized external economies or spillovers of externalities through inter-regional networks are more significant in improving productivity performance. Therefore, this study examined the relationships between productivity performance and geographical externalities from an industry perspective using regional economic data for Japan and employed a dynamic total factor productivity function approach. It also assessed the impact of high-quality transportation infrastructure improvements on productivity catch-up. The results revealed the long-run effects of inter-regional networks on the productivity performance of industries. Furthermore, it revealed that, as a geographical externality, the “borrowed size” effect contributed significantly to productivity improvements in both manufacturing and non-manufacturing industries. These findings suggest that enhancing both high-quality transportation infrastructure and inter-regional transportation networks is likely to improve industry productivity performance.

Keywords Inter-regional network · Borrowed size · Total factor productivity · Panel data · Japan

JEL Classification R10 · R11

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1 Introduction

This study explores the role of geographical externalities as determinants of the productivity performance of regional economies and statistically identifies their effects.

Advances in technology and networked societies have altered inter-regional exchange mechanisms and significantly affected the productivity performance of economic agents (Castells 1996). Economic agents in a region not only benefit from local, within-region interactions but also from external economies outside the region. Consequently, regional economies shift from being spatially independent areas to network hubs and region–region interactions affect the productivity performance of regional economies (Cooke et al. 2011).

Traditional arguments on economies of agglomeration have suggested that the geographical agglomeration of economic agents affects the productivity performance of the regional economy as external economies (Combes and Gobillon 2015; Rosenthal and Strange 2004). Several empirical studies (e.g., Acs et al. 2002; Carlino and Kerr 2015; Varga 1999) have shown that innovative activities responsible for productivity performance improvements tend to be spatially clustered and localized. Knowledge spillover depends on various factors, including tacit dimensions of knowledge, proximity of different types, and labor mobility (Grillitsch and Nilsson 2015). Tacit knowledge has been considered a localized factor, as knowledge transfer over long distances is difficult (Lam 2000).

However, non-geographically bound forms of knowledge spillovers exist, such as those based on cognitive, technological, and social proximities, through which several studies have indicated the possibility of knowledge exchange at a distance. For example, Asheim and Coenen (2006) and Trippel (2011) argue that acquiring localized knowledge is insufficient for innovation activities and that combining and complementing localized knowledge with knowledge acquired from distant sources is crucial. That is, collaboration between distant locations is considered an important factor for successful productivity performance improvement, and in this sense, traditional geographic proximity is neither a necessary nor a sufficient condition for productivity improvements (Boschma 2005).

The “borrowed size” concept—a theory on geographical externalities (Burger et al. 2015) proposed by William Alonso (1973)—theoretically explains the impact of such distant collaboration on the productivity performance of regional economies. According to this concept, regions adjacent to a central region can benefit from the latter’s external economies by borrowing the economies of scale generated therein (Meijers and Burger 2017). Inter-regional networks allow regions to acquire complementary capabilities through collaboration and benefit from economies of scale as external advantages from other regions (Capello 2000; Phelps et al. 2001). As a result, the economic growth of regions is influenced not only by the traditional significance of geographical closeness but also by their capacity to form connections with major urban centers (Meijers et al. 2016). Several studies (e.g., Camagni et al. 2015, 2016; Johansson and Quigley 2003; Melo et al. 2017; Otsuka 2018, 2020a, b, 2024a) have provided empirical evidence on these effects from a regional economic system perspective.

However, previous studies have not provided sufficient empirical evidence on regional industrial activity. Regional industries can be classified into export-oriented (core) and intra-regional (non-core) industries. In many cases, manufacturing industries represent export-oriented industries, which form an efficient production system in a wide-area network to meet demand from outside their regions. For example, the supply chains of advanced industries, such as the semiconductor sector, are constructed outside their regions, and inter-regional divisions of labor in production develop, likely improving inter-regional collaboration in industrial innovation activities. Intra-regional industries, however, include non-manufacturing industries, mainly service sectors. In these industries, the construction of production systems is limited to the region to efficiently respond to demand within the region. In other words, the production bases of these industries are localized; hence, the industries' innovation activities are likely to be localized as well. To clarify the effects of inter-regional networks that differ across industries, this study quantitatively evaluates the factors of geographical externalities that affect productivity performance using industry data.

Using Japanese industry-specific data, this study significantly contributes to evaluating regional industrial policy in Japan. In the 2000s, the period under observation, regional policies were continuously implemented to promote industrial location and strengthen inter-regional networks. In 2001, the Ministry of Economy, Trade and Industry (METI) launched the "Industrial Cluster Plan," followed by the Ministry of Education, Culture, Sports, Science and Technology (MEXT) initiating the "Knowledge Cluster Creation Project" in 2002. These initiatives were designed to enhance regional economic productivity by fostering connections among businesses, academic institutions, and research organizations across traditional administrative boundaries.

Such policies may improve productivity by enabling new technological and product developments through network formation, even in regions with scarce management resources. A research study conducted in France demonstrated that firms within competitive industrial clusters exhibited comparatively higher total factor productivity (TFP) than those outside such clusters (Abdesslem and Chiappini 2019). In other words, assuming Japan's regional policies have been as successful as policymakers anticipated, the overall productivity of the area may have improved, with less productive regions potentially catching up to their more economically advanced counterparts. Therefore, to verify the validity of these policies, it is crucial to examine the effects of inter-regional networks on productivity performance.

This study empirically analyzes geographical externalities by industry based on 2000–2018 data from 47 prefectures. The analysis method utilizes dynamic panel analysis based on dynamic TFP functions, allowing us to capture both short- and long-term effects of inter-regional networks on productivity performance. That is, it is possible to assess whether the formation of inter-regional networks has helped low-productivity regions to catch up with high-productivity regions over different timescales.

The structure of this paper is as follows. The connection between productivity performance and geographical externalities is explored in Sect. 2. Section 3 outlines the analytical approach, including the definition of the TFP indicator and a detailed

explanation of the analytical model and data. Section 4 discusses the findings from the estimation and provides an interpretation of the analytical outcomes. The study concludes in Sect. 5, summarizing the research and addressing unresolved issues.

2 Conceptualization

2.1 Productivity

Productivity is defined as the ratio of inputs to outputs. The higher the productivity, the more output is obtained from a given input, or the less input is required to produce a given amount of output. Generally, input is a production factor, such as labor or capital, and output is the value added.

Labor productivity, which is commonly used as a measure of productivity, is defined as the output per unit of labor. It is also simply referred to as factor productivity because it considers only labor as an input. This indicator is effective when the composition ratio of input goods is almost the same among economic agents. However, in regional economies, the composition ratio of input goods differs depending on the industrial structure of each region. In this case, labor productivity cannot accurately evaluate the productivity performance of the regional economy. For example, the labor productivity of economic agents that use capital-intensive technology tends to be higher; therefore, regions specializing in capital-intensive industries tend to have higher labor productivity. This raises the concern that it is not possible to distinguish whether this is because industries that use labor resources efficiently are concentrated in the region or because industries with a high capital–labor ratio are concentrated in the region.

In contrast, TFP is defined as the ratio of aggregate inputs to aggregate outputs and encompasses both labor productivity and capital productivity. Therefore, it does not face concerns that arise with factor productivity (Otsuka and Goto 2016).

Table 1 shows the TFP content. TFP consists of direct technological progress, such as research and development (R&D), knowledge stocks, knowledge spillovers,

Table 1 Contents of TFP

Types of technological progress	Contents
Direct technological progress	R&D investment, Knowledge stock, Knowledge spillover
Indirect technological progress	Economies of scale Capital maturity improvement Human capital effects Infrastructure Geographical externalities (agglomeration and network economies)

Watanabe (2001) and Otsuka (2008)

and indirect technological progress, such as in external economies (Watanabe 2001; Otsuka 2008).

Table 1

Direct technological progress mainly involves R&D investment by economic agents. R&D investment and the stock of knowledge directly improve the productivity of economic agents. In addition, knowledge spillovers, in which some knowledge of other economic agents contributes to technological progress, also improve the productivity performance of economic agents.

Indirect technological progress includes technological progress embodied in capital and labor. For example, when old capital stock is replaced with new capital stock through capital investment, even if the total amount of capital stock remains the same, the new capital stock can potentially produce more output than the old capital stock due to the addition of new functions and the increase in capital expertise. Similarly, when a more skilled worker replaces one worker, more outputs can be generated even if the number of workers remains the same. In addition, if economies of scale exist, productivity performance will improve due to the benefits of mass production. Productivity performance will also improve if managerial efficiency increases—that is, if operational efficiency improves due to a review and acceleration of the decision-making process. Furthermore, if industries are geographically connected, geographical externalities will be strengthened due to agglomeration economies or inter-regional network economies. Consequently, productivity performance in the regional economy will improve.

2.2 Borrowed size

The concept of borrowed size integrates population, function, and critical mass effects for both demand and supply, proposing that smaller urban areas can benefit from nearby larger agglomerations without experiencing their drawbacks, such as congestion. This theory suggests that the advantages of urban clustering are not confined to a city's physical limits but can extend to surrounding regions. This framework explains why smaller cities might operate more efficiently than their larger counterparts and why rural areas can maintain effective urban structures by “borrowing” agglomeration economies from the broader urban network (Parr 2002).

As urban agglomeration externalities broaden their reach, their influence extends beyond traditional city limits. This expansion suggests that the notion of borrowed size could potentially elucidate the growth of regional economies on a larger scale, encompassing areas well outside urban centers. That is, the borrowed size concept can overcome the limitations of the conventional geographical approach that only cities adjacent to a metropolis can experience increased efficiency, and argues that scale is not the only determinant of productivity and economies of agglomeration. Incorporating long-distance cooperation agreements throughout the urban network, along with the presence of advanced functions, could increase urban productivity despite the limited size of cities (Boix and Trullén 2007; Camagni 1993; Capello 2000; Camagni and Capello 2004; Hall and Pain 2006; Huang et al. 2020).

Establishing efficient transportation systems enhances these borrowed-size effects by reducing travel expenses for innovators and expanding opportunities for economic interactions. A well-developed transportation infrastructure not only boosts the efficiency of its direct users but also benefits related parties by accelerating and increasing the frequency of information exchange. This means that the geographical reach of externalities grows as knowledge spreads beyond regional boundaries. Consequently, creating an advanced transportation network amplifies network externalities by enabling innovators to move more freely between the regions (Ma and Liu 2022; Tian et al. 2022).

2.3 Agglomeration shadow

Nevertheless, some scholars contend that establishing regional networks may not always benefit urban development (Hesse 2016). The rivalry among cities that emerges from these networks can result in a “siphoning effect,” where larger urban centers extract resources from their smaller neighbors, impeding the growth of these surrounding cities. Consequently, this inter-urban competition may decrease the overall efficiency of urban development processes (Dijkstra et al. 2013).

Fujita et al. (1999) theoretically explained this phenomenon as the “agglomeration shadow.” An agglomeration shadow is a phenomenon in which large cities and regions with strong economic agglomeration have an economic impact on neighboring smaller regions (Fujita and Thisse 2002). Specifically, it refers to a situation in which the economic activities of small cities and suburbs near large cities are suppressed or inhibited from growth due to large cities absorbing resources, including human resources, from those neighboring regions. The agglomeration shadow effect and the borrowed size effect are related and affect the economic dynamics between regions (Meijers 2005; Phelps and Ozawa 2003; Burger and Meijers 2016).

The two concepts may have a complementary relationship. Small cities and suburban areas located near large cities can be overwhelmed by the economic activities of large cities, but they can also grow using the resources and infrastructure of the large cities. In other words, while some economic activities are suppressed, others can benefit.

However, it is also possible that a competitive relationship will arise between the two concepts. On the one hand, if the borrowed size effect is fully demonstrated, the impact of the agglomeration shadow will weaken, and the surrounding areas may achieve more independent economic growth. On the other hand, if the agglomeration shadow has a strong effect, the borrowed size effect will be canceled out, and the growth of the surrounding areas will be severely restricted.

These relationships are an important point of discussion in formulating regional policies and urban planning; however, as discussed by Hornbeck et al. (2004), the circumstances under which the agglomeration shadow becomes dominant have not been fully clarified owing to the complexity of the analytical approach. Nevertheless, a complex relationship exists between the borrowed size effect and agglomeration shadow. In other words, the borrowed size effect can be strengthened or weakened by the agglomeration shadow. To avoid the difficulty of interpreting agglomeration shadows, this

study does not explicitly consider agglomeration shadows but instead focuses on the net effect of geographical externalities from a regional macroperspective and statistically examines the impact of the borrowed size effect on the productivity performance of regional economies.

2.4 Research focus

This study statistically clarifies the effects of inter-regional network formation on the productivity performance of regional economies relative to conventional geographical externalities.

Research has shown conflicting results regarding the relationship between inter-regional network externalities and traditional geographical externalities. Some studies indicate that these two types of externalities complement each other (Boix and Trullén 2007; Tian et al. 2022; Wetwitoo and Kato 2017). Conversely, other research suggests that network externalities can substitute conventional geographical externalities (Fu et al. 2023; Huang et al. 2020; McCann and Acs 2011; Tang et al. 2021). Notably, recent investigations in China have provided substantial evidence that network externalities exert a greater influence on innovation activities within special economic zones compared to traditional geographical externalities (Jiang et al. 2022; Liu et al. 2023; Yang et al. 2022; Zheng et al. 2023).

To contribute to the discussion of previous research, this study clarifies the effects of borrowed size on productivity performance improvements from an industrial perspective. In particular, using a dynamic production function approach clarifies the short- and long-term effects of geographical network economies on the productivity performance of industries.

3 Methods

3.1 TFP index

This study calculates TFP using the methodologies outlined by Good et al. (1997), Aw et al. (2001), and Otsuka (2018, 2024a; b). The approach measures an individual economic agent's TFP as a relative deviation from a representative economic agent, which is defined as having average input and output levels at a specific point in time. By accounting for changes in the representative agent's TFP over time, this method enables the comparison of an individual economic agent's TFP across different periods.

Let us assume a particular industrial sector. In this case, the relative TFP level of region j ($j = 1, \dots, N$) for time t ($t = 1, \dots, P$) or $\ln TFP_{jt}$ is expressed as follows:

$$\ln TFP_{jt} = \left(\ln Y_{jt} - \overline{\ln Y_t} \right) + \sum_{s=1}^t \left(\overline{\ln Y_s} - \overline{\ln Y_{s-1}} \right) - \left[\sum_{i=1}^n \frac{1}{2} \left(S_{ijt} + \overline{S_{it}} \right) \left(\ln X_{ijt} - \overline{\ln X_{it}} \right) + \sum_{s=1}^t \sum_{i=1}^n \frac{1}{2} \left(\overline{S_{is}} + \overline{S_{is-1}} \right) \left(\overline{\ln X_{is}} - \overline{\ln X_{is-1}} \right) \right]. \quad (1)$$

In this equation, Y_{jt} represents the output of region j at time t , while X_{ijt} denotes the input factor i ($i = 1, \dots, n$) in region j at time t . In addition, S_{ijt} signifies the cost share of input factor i in region j at time t . The superscript lines positioned above each symbol represent the average of each variable across regions. Equation (1) demonstrates that regional TFP levels can be evaluated in terms of relative TFP levels at specific times and locations and by considering TFP level changes over time.

3.2 Determinants model on TFP index

Equation (2) is assumed to be the conventional panel model used to evaluate the effect of geographical externalities on TFP performance:

$$\ln TFP_{jt} = b_{BS} \ln BS_{jt} + b_{DS} \ln DS_{jt} + b_{TC} \ln TC_{jt} + b_{HC} \ln HC_{jt} + c_j + u_{jt}, \quad (2)$$

where all variables are logarithmic: j is the region and t is the time, TFP is the TFP index defined in Eq. (1), the BS variable is an index capturing the effect of borrowed size, and the DS variable measures population density. The TC variable represents the traffic capital stock, the HC variable measures the pool of skilled laborers, c is a constant term on individual effects, and u is the error term. In the variables for geographical externalities explaining TFP performance, the BS variable complements externalities that spill over across regions, while the non-BS variables complement localized externalities.

The definitions of each determinant with respect to geographical externalities are explained below.

First, the BS variable is defined as follows:

$$BS_{jt} \equiv \sum_{k \neq j} \frac{GRP_{kt}}{d_{jkt}},$$

where GRP is the regional value added, and d_{jkt} is the distance resistance (travel time distance). This variable is characterized by its unique approach to measuring cluster proximity, which considers cluster size and distance resistance. The variable's magnitude is influenced by the borrowed size from clusters in other regions, factoring in the distance resistance to these clusters. As clusters in other regions grow larger, the corresponding variable increases in value. In addition, the BS variable's value rises when the distance resistance to clusters in other regions decreases.

GRP is a proxy variable for cluster size. Previous studies have used population size and density as proxy variables for cluster size (Camagni et al. 2015, 2016). For Japan, however, there are concerns about following the methods employed in previous studies. For example, many residents of Kanagawa Prefecture, which is adjacent to Tokyo, commute to Tokyo. In this case, the population size is large relative to the economic size of Kanagawa Prefecture, resulting in an overestimation of the BS variable. Therefore, GRP is preferable to population size or density as an indicator that adequately expresses the economic scale.

Distance resistance, which constitutes the BS variable, is the travel time distance between the regions. Previous studies have used physical distance as

distance resistance (Hansen 1959; Camagni et al. 2015, 2016). However, travel time distance should be preferable to physical distance to accurately measure the magnitude of traffic between the regions in a business. This is because remote regions with a well-developed, high-quality transportation network can access the core region more quickly.

The calculation of distance resistance d involves a two-step process. Initially, the proportion of passengers using each transportation mode (air, rail, and car) is determined for every possible origin–destination pair. Subsequently, the distance resistance is computed by calculating a weighted average of the travel time for each mode, using the previously determined proportions as weights. This calculation is represented by a mathematical equation as follows:

$$d_{jkt} = A_{jkt} \times \text{air travel time} + B_{jkt} \times \text{rail travel time} + C_{jkt} \times \text{car travel time},$$

Where A_{jkt} , B_{jkt} , and C_{jkt} are the transportation shares in year t , with region j as the origin and region k as the destination; A denotes air, B denotes rail, and C denotes car.

Population density, represented by the DS variable, serves as a proxy for measuring economies of density. This variable is calculated by dividing the population by the habitable land area. Research by Ciccone and Hall (1996) and Ciccone (2002) indicates a positive correlation between density and productivity, referred to as economies of density. Rosenthal and Strange (2004) found that a twofold increase in employment density leads to a 5% to 6% rise in labor productivity, attributable to these density-related economic benefits.

The TC variable represents the aggregate of transportation-related social infrastructure, encompassing the social capital stock of roads, ports, and airports. Firms in regions with superior transportation infrastructure can benefit from reduced travel times, lower transportation expenses, and increased shipping capacities (Aschauer 1989). This infrastructure development enhances industries' productive potential, allowing them to source inputs more efficiently and optimize their production facilities, decreasing production costs. Research has demonstrated that areas with high-quality social capital stocks exhibit greater TFP or the Solow residual (Bröcker et al. 2019; Eberts and McMilen 1999; Otsuka and Goto 2015; Otsuka 2017).

The HC variable is an indicator representing the pool of skilled laborers. Since no official statistics exist, the indicator is created by defining it using the following equation:

$$HC_{jt} \equiv \frac{\sum_n \sum_s w_{snt} \times L_{snt}}{L_{jt}},$$

where j is the region, t is the time, n is the education level (universities, junior colleges, high school), and s is the gender (male or female). w is the wage-to-average wage ratio by education level. L is the number of laborers by education. The wages of laborers with a university degree or higher are relatively high compared to the average wage. Therefore, the HC variable is higher in regions where the number of

laborers with a college degree or higher is higher, and the HC variable is lower in regions with many laborers with a high school education or lower.

Conventional economic theory recognizes that educational investments enhance workers' human capital and boost productivity. The link between human capital and TFP growth has been extensively examined by scholars such as Schultz (1961), Becker (1964), Welch (1970), and Maudos et al. (1999). Investing in human capital increases employment and per capita income by enabling the workforce to effectively utilize knowledge, subsequently facilitating knowledge dissemination (Huang et al. 2019). Consequently, human capital significantly and enduringly influences TFP (Ghosh and Parab 2021; Habib et al. 2019).

3.3 Estimation model

There are endogeneity concerns with the estimation of Eq. (2) (Graham 2009). In the field of geographical externalities, researchers often assume that the error terms in the equation are independent of the regression coefficients. However, this assumption becomes invalid when dealing with endogenous regression coefficients. The factors involved in geographical externalities are linked to productivity and may themselves be endogenous variables.

Many previous studies have addressed this concern through two-step least squares methods with instrumental variables. However, some have reported that the estimates are unlikely to be biased even if geographical externalities are endogenous. For example, Ciccone (2002) and Ciccone and Hall (1996) have shown that estimating geographical externalities with instrumental variables slightly differs from estimating them with the least squares method. Studies by Henderson (2003) and Rosenthal and Strange (2004) have also determined that endogenous regression coefficients have a minimal effect. However, Combes and Gobillon (2015) have pointed out that the issue should be effectively addressed even though solid evidence that bias arises from endogeneity is lacking.

In this study, we address this concern using dynamic panel estimation, in which lags of the explained variable are introduced to the explanatory variables (Arellano and Bond 1991). This method addresses the concern that the explanatory variables and the error terms will be correlated and inconsistent in the usual ordinary least squares estimation (Cameron and Trivedi 2009). This analytical approach is well suited for numerous individual effects and limited time-series data scenarios. It particularly applies to our research, given that our dataset encompasses a relatively brief time frame (18 years) and a wide array of cross-sectional units (47 in total).

The estimation model for dynamic panel estimation for Eq. (2) is as follows:

$$\Delta \ln TFP_{jt} = \alpha \Delta \ln TFP_{jt-1} + \beta \Delta \ln BS_{jt} + \gamma \Delta \ln DS_{jt} + \delta \Delta \ln TC_{jt} + \lambda \Delta \ln HC_{jt} + \Delta \varepsilon_{jt}, \quad (3)$$

where Δ represents the difference. The Greek letters of the regression coefficients in the equation are the estimated parameters.

If the effect of borrowed size is significant, the sign of β is positive. When the effect of population density is manifest, the sign of γ is positive, but when the

negative external effect associated with overcrowding is manifest, the sign of γ is negative. The sign of δ is positive because traffic capital, as a geographical externality, raises the productivity of the industry. However, when negative external effects such as traffic congestion are large, the sign of δ is negative because transportation capital overcrowding reduces the industry's productivity. The sign of λ is positive because an increase in the educational capacity of laborers increases the industry's productivity performance.

Short- and long-run elasticities are obtained from the estimated results of Eq. (3). We assume that short-run TFP is less responsive to geographical externalities than long-run TFP because, while some changes in the production system, such as the operation of plants and equipment, can be adjusted immediately, others, such as adjustments in the amount of employment or location, cannot. That is, we cannot expect the production system to change quickly in response to geographical externalities. Therefore, short-term output may deviate from the optimal output in the long run. The production system will not adjust immediately to the long-run equilibrium but should converge to the optimal level over time if producers are rational.

Traditional TFP functions typically omit the lag variable for TFP in their estimations due to the absence of an adjustment process for productivity performance. This approach assumes that production at different time points is unrelated, resulting in no lag in the adjustment process. The standard TFP function presumes that expectations do not incur costs that influence current decisions, which is not a realistic assumption. In contrast, the dynamic TFP function model incorporates the adjustment process of productivity performance, thus including the lag variable of TFP in its estimation. This model allows for a gradual adjustment between the optimal long-term productivity level and the short-term productivity level. Consequently, researchers can derive both short-term and long-term elasticities by utilizing the dynamic TFP function model, such as Eq. (3).

3.4 Data

The study employs panel data from 2000 to 2018, examining all industries, manufacturing and non-manufacturing industries. The analysis covers 47 prefectures, as illustrated in Fig. 1. All original data in this study are accessible to the public and can be obtained at no cost.

To measure TFP, the study utilizes the Regional-Level Japan Industrial Productivity Database (R-JIP), which is published by the Research Institute of Economy, Trade and Industry (RIETI). In Japan, few TFP analyses target industries (e.g., Goto et al. 2018) because the data for measuring TFP have not been well developed. By employing a database that has been newly compiled by the RIETI, we can measure TFP by industry from 2000 to 2018. In our analysis, we employ real value added as the output (Y) while utilizing labor and capital input as the inputs (X). Labor input is quantified through labor hours, and capital input is measured using real capital stock. To calculate the cost share of inputs (S), we rely on the labor and capital costs recorded in the database.

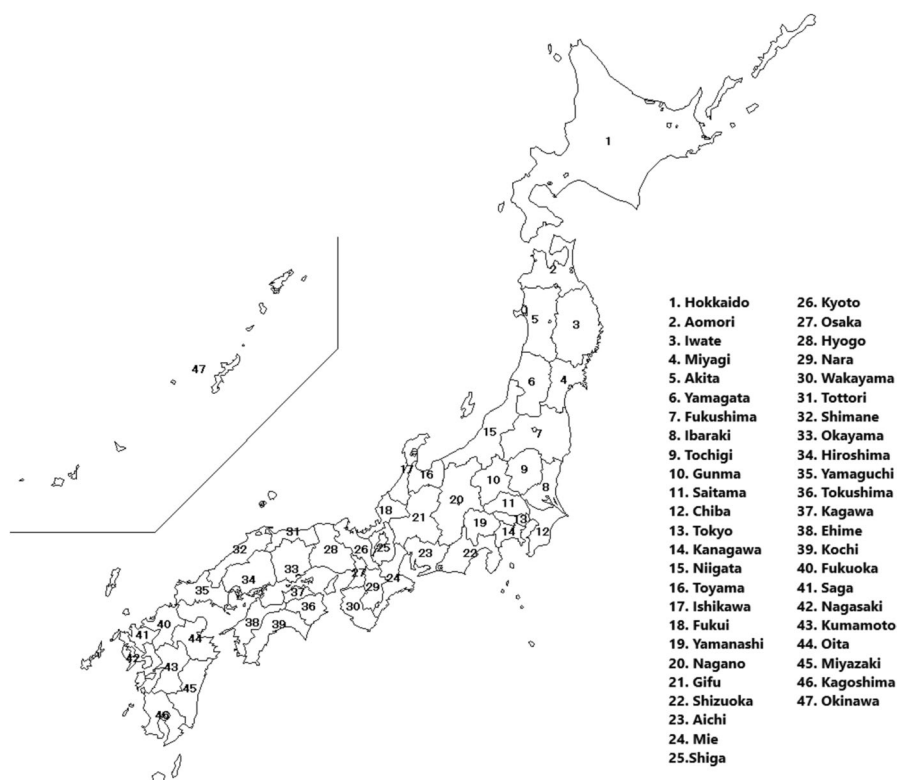


Fig. 1 47 Prefectures in Japan. Otsuka (2021)

Data for the BS variables, specifically time–distance data, are sourced from the National Integrated Traffic Analysis System (NITAS), which is managed by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT). Meanwhile, the DS variables utilize population and area statistics from the Social and Demographic Statistics System, operated by the Statistics Bureau within the Ministry of Internal Affairs and Communications. Data on social capital stock related to transportation are from the Cabinet Office. The data on the number of laborers by education and wages by education used to create the variables related to the pool of skilled laborers are from the Basic Survey on Employment Structure (Statistics Bureau, Ministry of Internal Affairs and Communications) and the Basic Survey on Wage Structure (Ministry of Health, Labour and Welfare).

Descriptive statistics, including the mean, standard deviation, highest and lowest values, and the range, are presented in Table 2 at decade-long intervals for the dataset. The average TFP for all industries was 0.035 in the 2000s and 0.075 in the 2010s, indicating an increase throughout the observation period. The range—the difference between the maximum and minimum values of TFP by region—is 0.727 for the 2000s and 0.761 for the 2010s, showing that TFP for all industries in Japan was increasing while simultaneously widening regional disparities.

Table 2 Descriptive statistics

	TFP index			BS (million yen)	DS (person/ha)	TC (million yen)	HC (index)
	ALL	MANU	NMANU				
2000–09							
Mean	0.035	0.179	– 0.007	128,885,317	13.714	4,878,493	0.744
S.D	0.124	0.216	0.118	32,576,330	16.574	3,547,188	0.057
Max	0.545	0.866	0.579	236,486,388	93.436	19,414,919	0.895
Min	– 0.182	– 0.436	– 0.188	71,952,719	2.522	1,612,596	0.603
Range	0.727	1.302	0.767	164,533,669	90.914	17,802,323	0.292
2010–18							
Mean	0.075	0.366	– 0.009	129,044,564	13.642	5,221,596	0.812
S.D	0.136	0.182	0.134	32,707,597	17.353	3,713,347	0.043
Max	0.600	0.902	0.619	241,103,250	97.697	20,199,760	0.934
Min	– 0.160	– 0.246	– 0.212	68,708,207	2.366	1,923,977	0.681
Range	0.761	1.149	0.831	172,395,043	95.332	18,275,783	0.253
2000–18							
Mean	0.054	0.267	– 0.008	128,960,750	13.680	5,041,015	0.776
S.D	0.131	0.221	0.126	32,638,672	16.948	3,630,888	0.061
Max	0.600	0.902	0.619	241,103,250	97.697	20,199,760	0.934
Min	– 0.182	– 0.436	– 0.212	68,708,207	2.366	1,612,596	0.603
Range	0.783	1.338	0.831	172,395,043	95.332	18,587,164	0.331

ALL all industries, MANU manufacturing industries, and NMANU represents non-manufacturing industries

The TFP trends across industries reveal that the manufacturing industry has consistently outperformed the non-manufacturing industry in terms of productivity. The manufacturing industry experienced continuous growth in TFP throughout the study period, whereas the non-manufacturing industry showed minimal improvement. In addition, the range of TFP in the manufacturing industry decreased over time, suggesting a reduction in regional disparities. In contrast, the non-manufacturing industry's TFP range expanded, indicating an increase in regional productivity disparities throughout the observation period.

In addition, Table 2 displays descriptive statistics for variables related to geographical externalities, which influence industry-based TFP. The BS variable increased throughout the observation period. While its maximum value increased, its minimum value decreased, suggesting that regional disparities in the borrowed size may have widened. The DS variable did not change significantly throughout the observation period, but its maximum value increased significantly, indicating that regional disparities were widening. Both the TC and HC variables increased throughout the observation period. However, the difference between the maximum and minimum values of the TC variable is widening, while the range of the HC variable is shrinking. This suggests that improvements in laborers' skills are progressing nationwide due to the increasing number of highly educated laborers.

4 Results and discussion

The estimation results for Eq. (3) are presented in Table 3. The table displays data for three categories: ALL, encompassing all industries; MANU, representing the manufacturing industry; and NMANU, denoting the non-manufacturing industry. When performing dynamic panel estimation, ensuring that the error terms do not exhibit serial correlation is crucial. A reliable estimator can only be achieved when this requirement is met. Therefore, an Arellano–Bond test must be performed, and while the first-order autocorrelations are allowed to be significant, the second-order autocorrelations are not. The results show that the estimation satisfies this condition for all models. Furthermore, we conduct a Sargan–Hansen test, an exogenous test for instrumental variables; the results indicate that the number of instrumental variables is not excessive, so the conditions for dynamic panel estimation are satisfied. The calculated coefficients represent elasticities since both the independent and dependent variables are expressed in logarithmic form. Consequently, a higher estimated coefficient value indicates a stronger influence of the independent variable on the dependent variable.

We first evaluate the estimation results for all industries and find that the signs of all regression coefficients are positive, and all variables related to geographical externalities positively affect TFP performance; the BS variable is statistically

Table 3 TFP regression in Japan (2000–2018)

Dependent variable: $\Delta \ln \text{TFP}(t)$

	ALL	MANU	NMANU
Regressors			
$\Delta \ln \text{TFP}(t-1)$	0.424 ** (0.034)	0.446** (0.018)	0.381** (0.051)
$\Delta \ln \text{BS}$	0.153 * (0.066)	0.190* (0.085)	0.169** (0.022)
$\Delta \ln \text{DS}$	0.402 ** (0.048)	– 0.015 (0.152)	0.151* (0.061)
$\Delta \ln \text{TC}$	0.244** (0.037)	0.909** (0.031)	– 0.130** (0.015)
$\Delta \ln \text{HC}$	0.187 ** (0.001)	0.132** (0.039)	0.130** (0.016)
Number of observations	799	799	799
<i>J</i> -Statistic	44.458	44.413	41.100
Prob (<i>J</i> -statistic)	0.369	0.370	0.157
<i>m</i> -Statistic (AR(1))	– 5.098 **	– 3.039**	– 3.987**
<i>m</i> -Statistic (AR(2))	– 0.510	1.025	– 1.011
Instruments	$\Delta \text{TFP}(t-2)$, One period lag of explanatory variables		

The regressions are estimated using panel data for all prefectures

ALL all industries, MANU manufacturing industries, NMANU represents non-manufacturing industries

The two-step first-difference GMM estimation method is used, and the standard errors are given in parentheses under the coefficients. The statistical significance of the coefficients is noted at either the 1% (**) or 5% (*) level. The *J*-statistic results are obtained from a Sargan–Hansen test of the over-identifying restrictions for the two-step GMM estimators. The *m*-statistics refer to tests for first- and second-order serial correlations

significant, and externalities due to inter-regional networks have a positive impact on TFP performance. The DS, TC, and HC variables, which represent localized externalities, are also statistically significant. In particular, the regression coefficient for the DS variable is the largest, indicating that population density strongly influences TFP. The TC and HC variables also significantly exceed the borrowed size effect, indicating that localized externalities are larger than inter-regional spillovers on productivity performance for all industries and that each relationship is complementary. These results support those of previous studies (i.e., Boix and Trullén 2007; Tian et al. 2022; Wetwitoon and Kato 2017), suggesting complementary relationships between network and localized externalities.

Next, we evaluate the estimation results for two industries and find that, on the one hand, the BS variable is positive and statistically significant for both the manufacturing and non-manufacturing industries. Further, the regression coefficients are larger for the manufacturing industry than for the non-manufacturing industry, indicating that the effect of borrowed size on TFP is larger for the manufacturing industry. On the other hand, the effects of population density and traffic capital stock differ significantly between the manufacturing and non-manufacturing industries; the regression coefficients for the DS variable are statistically significant only for the non-manufacturing industry. This result suggests that the effect of population agglomeration on TFP performance is stronger for the non-manufacturing industry, supporting the fact that the non-manufacturing industry is an intra-regional industry. The effect of traffic capital stock is large for the manufacturing industry and positively affects TFP; since the value of the regression coefficient for the TC variable is the largest among the explanatory variables, it is inferred that the enriched transportation infrastructure has had the greatest influence in the manufacturing industry's TFP performance. The effect on the pool of skilled laborers is positive for both manufacturing and non-manufacturing industries, but the magnitude of the regression coefficient on the HC variable is small, and its effect on TFP performance is the least significant among the explanatory variables.

Table 4 shows the magnitude of the short- and long-run elasticities to TFP for each variable, calculated based on the estimates in Table 3. For all variables, long-run elasticities exceed short-run ones. For example, the impact of the BS variable is greatest in the long run for the manufacturing industry. This indicates that the impact of inter-regional network formation on TFP is greater in the manufacturing industry in the long run. In particular, the long-term impact of the TC variable on TFP improvement is

Table 4 Elasticities

Variables	ALL		MANU		NMANU	
	Short-run	Long-run	Short-run	Long-run	Short-run	Long-run
BS	0.153	0.265	0.190	0.343	0.169	0.274
DS	0.402	0.697	− 0.015	− 0.028	0.151	0.244
TC	0.244	0.424	0.909	1.639	− 0.130	− 0.210
HC	0.187	0.324	0.132	0.237	0.130	0.209

ALL all industries, MANU manufacturing industries, NMANU non-manufacturing industries

substantial. This result indicates that developing high-quality transportation infrastructure is likely to significantly improve the productivity performance of the manufacturing industry through enhanced intra-regional and inter-regional externalities. On the other hand, the DS variable has a positive long-run impact only for the non-manufacturing industry. This suggests that the non-manufacturing industry is likely to benefit significantly from the size effect of local demand as the increase in population agglomeration results in more efficient sales channel formation.

The estimation results for a restricted observation period are presented in Table 5, demonstrating the robustness of the findings. Both the serial correlation and Sargan–Hansen tests yield results that meet the necessary criteria for assessing dynamic panel estimation. None of the models shows significant changes in the relative magnitude of the regression coefficients. Furthermore, the statistical significance of the regression coefficients does not show significant change. In other words, the results of the models in this study are stable.

Finally, using the estimation results in Table 3, the contribution of the BS variable to TFP improvement was evaluated based on the following equation:

$$\text{Contribution}_j \equiv \frac{\beta}{1 - \alpha} \times \Delta \ln BS_{j,2018-2000}$$

where $\beta/(1 - \alpha)$ denotes the long-run elasticities on the BS variable.

Table 5 TFP regression in Japan (2000–2015)

Dependent variable: $\Delta \ln \text{TFP}(t)$

	ALL	MANU	NMANU
Regressors			
$\Delta \ln \text{TFP}(t-1)$	0.294** (0.033)	0.370** (0.015)	0.401** (0.059)
$\Delta \ln \text{BS}$	0.248** (0.075)	0.401** (0.060)	0.212** (0.027)
$\Delta \ln \text{DS}$	0.398** (0.036)	– 0.344 (0.152)	0.227* (0.094)
$\Delta \ln \text{TC}$	0.320** (0.039)	1.151** (0.023)	– 0.100** (0.023)
$\Delta \ln \text{HC}$	0.224** (0.056)	0.197** (0.037)	0.146** (0.020)
Number of observations	658	658	658
<i>J</i> -Statistic	44.000	45.060	34.051
Prob (<i>J</i> -statistic)	0.387	0.345	0.165
<i>m</i> -Statistic (AR(1))	– 4.900**	– 1.751**	– 3.891**
<i>m</i> -Statistic (AR(2))	– 1.350	0.581	– 0.935
Instruments	$\Delta \text{TFP}(t-2)$, One period lag of explanatory variables		

The regressions are estimated using panel data for all prefectures

ALL all industries, *MANU* manufacturing industries, *NMANU* represents non-manufacturing industries. The two-step first-difference GMM estimation method is used, and the standard errors are given in parentheses under the coefficients

The statistical significance of the coefficients is noted at either the 1% (**) or 5% (*) level. The *J*-statistic results are obtained from a Sargan–Hansen test of the over-identifying restrictions for the two-step GMM estimators. The *m*-statistics refer to tests for first- and second-order serial correlations

Figures 1, 2 and 3 show the contribution of the BS variables from 2000 to 2018 on the vertical axis and TFP in 2000 on the horizontal axis. Figure 2 shows the results for all industries, Fig. 3 for manufacturing, and Fig. 4 for non-manufacturing. If the relationship between the two variables in each figure is negative, i.e., the relationship is declining to the right, it means that the contribution of the BS variable was higher in regions where productivity was relatively low at the beginning of the period. In other words, the BS variable reduced the TFP disparity throughout the observation period. In all figures, a clear rightward relationship is observed.

Table 6 shows the correlation test results for the two variables in Figs. 2, 3 and 4. In all the cases, the correlation coefficients are negative and statistically significant. This result indicates that, in all cases, the BS variable may have reduced regional disparities in TFP.

Table 6

In Japan, Shinkansen lines (bullet train) were extended in the Tohoku and Kyushu regions in the 2000s, and the Kyushu Shinkansen line was fully opened in the 2010s, significantly reducing the extended regions' travel time for accessing large metropolitan areas. Therefore, it is highly likely that the productivity performance of industries located in the Tohoku and Kyushu regions, which had low productivity in the early 2000s, has improved significantly due to the development of high-quality transportation. The results of this study suggest that developing a high-quality

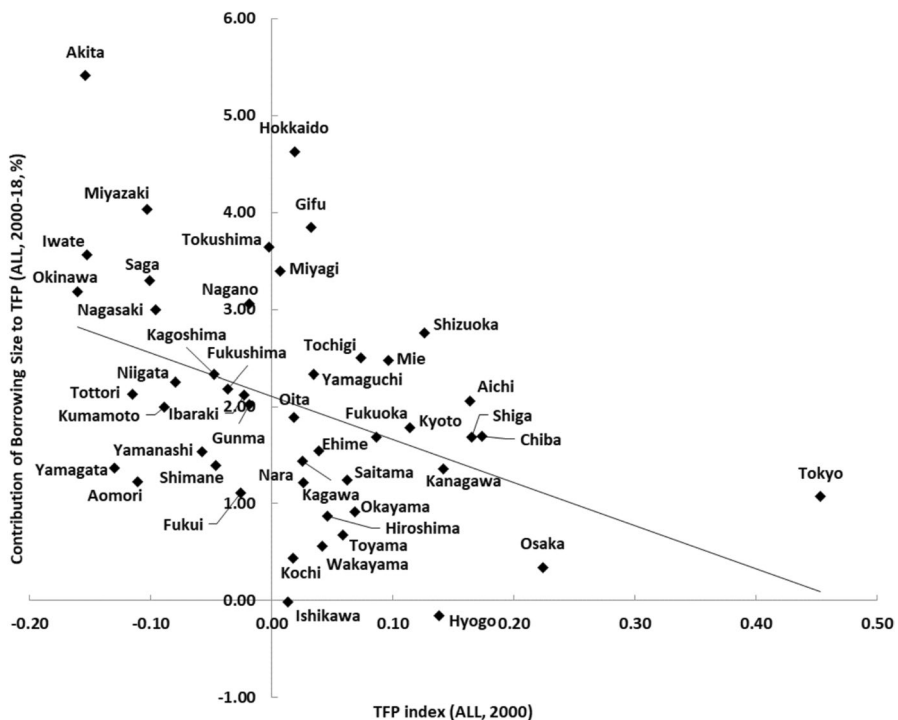


Fig. 2 Relationship between contribution of the BS index and initial TFP. (All industries)

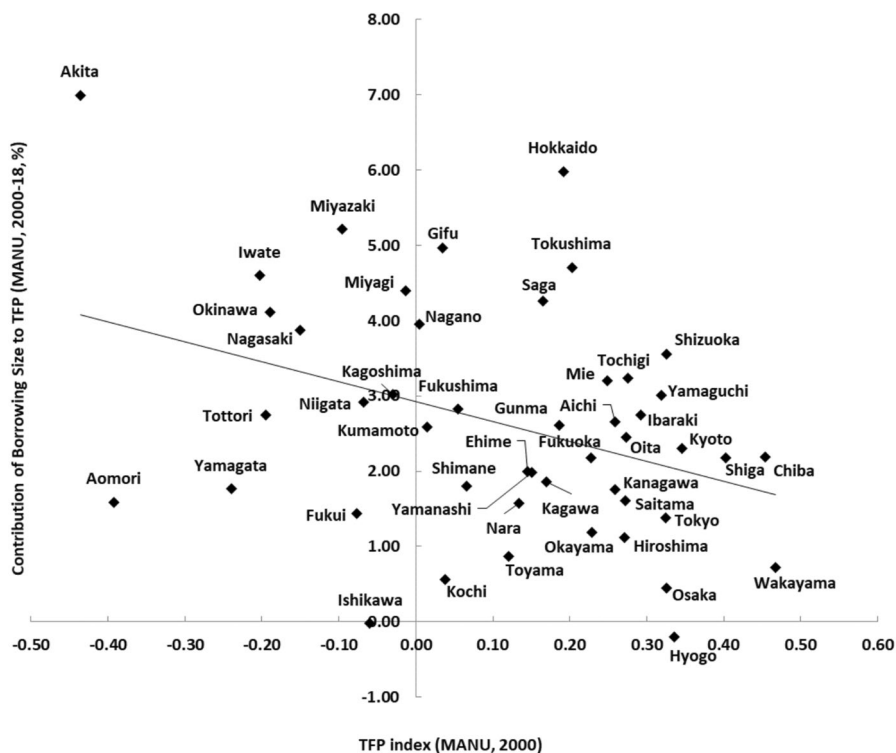


Fig. 3 Relationship between contribution of the BS index and initial TFP. (Manufacturing)

transportation network may significantly contribute to improving industries' productivity performance.

5 Conclusion

To assess the effect of inter-regional networks on industries' productivity performance, this study statistically evaluates geographical externalities affecting TFP using a dynamic TFP function approach. In particular, this study examines the importance of localized and inter-regional spillover externalities in determining Industry-based TFP. The results indicate that the borrowed size effect influences the TFP of both manufacturing and non-manufacturing industries. In particular, the long-run effect of borrowed size was larger for manufacturing industries, indicating that geographical externalities interact across regional boundaries in the long run as high-quality transportation networks expand the geographic scope of production activities.

The study also finds that the contribution of borrowed size effects to TFP improvement differs significantly across regions, with the impact of these effects being greater in less productive regions than in more productive ones. In other words, low-productivity regions may improve their TFP by connecting to

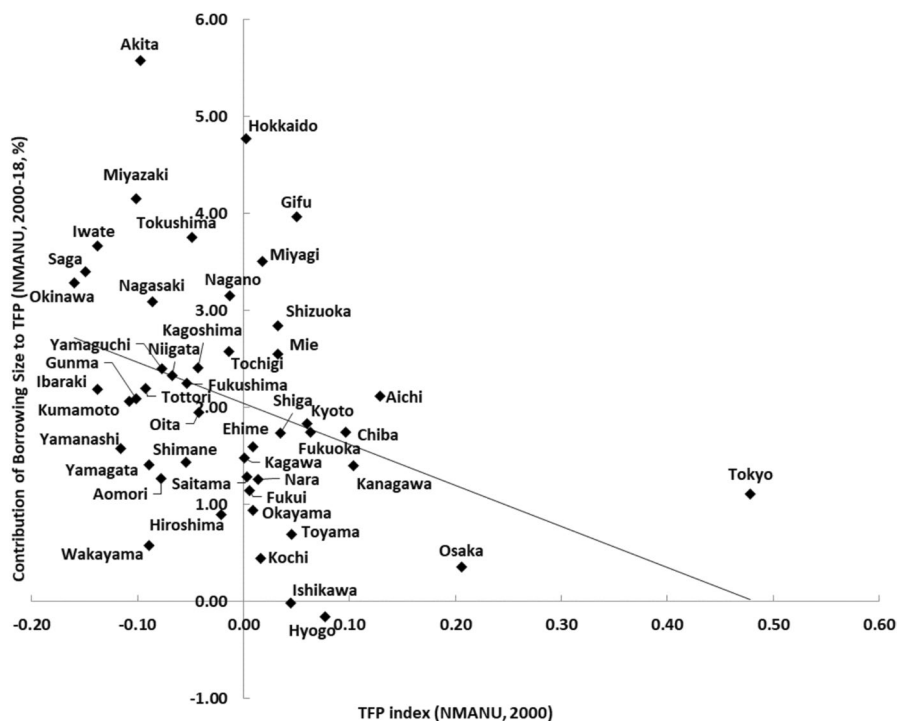


Fig. 4 Relationship between contribution of the BS index and initial TFP. (Non-manufacturing)

Table 6 Correlation statistics in Figs. 1, 2, and 3

	ALL	MANU	NMANU
Correlation coefficient	− 0.427	− 0.368	− 0.369
<i>t</i> -value	3.164	2.657	2.663
<i>p</i> -value	0.003	0.011	0.011

ALL represents all industries, *MANU* represents manufacturing industries, *NMANU* represents non-manufacturing industries

inter-regional networks and strengthening their relationships with other regions, allowing the externalities of other regions to spill over inward through increased opportunities for interactions among regions.

Furthermore, this study confirms the importance of localized externalities for both manufacturing and non-manufacturing industries. The impact of localized external economies on TFP performance is particularly large in the non-manufacturing industry; therefore, this reflects how the non-manufacturing market is limited within the region. Thus, we conclude that the borrowed size effect complements the localized external economy within the region.

The results of this study suggest that to effectively improve the productivity performance of industries, both strategically strengthening individual regions and

developing a high-quality transportation network that connects regions are essential. Connecting clusters with a high-quality rapid transit network and strengthening inter-regional linkages will result in significant between-region externality spillovers, which were previously region-bound. A high-quality transportation network can expand externalities over a wider area through enhanced borrowed size effects.

Furthermore, many developed countries aim to enhance economic sustainability by promoting regional innovation. If the desired growth strategy is formulated from the perspective of the location environment, the development of skilled laborers will likely increase industrial competitiveness across all industries. The results of this study confirm that improving labor skills contributes to improving the productivity performance of industries. Policy support for regional universities and research institutes can effectively increase an industry's productivity performance.

While this study provides unique insights, we could not fully discuss the possibility of spatial autocorrelation; nevertheless, spatial interactions exist in the production activities of economic agents (Rice et al. 2006). A credible, dynamic panel estimation that considers spatial autocorrelation has not been fully developed, making it difficult to statistically address the possibility of spatial autocorrelation in this study. This issue should be addressed by developing analytical methods in the future.

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Data availability The datasets used and/or analyzed during the current study are available from the corresponding author on request. Data Source: The data utilized in this study is accessible to the public at the following web address: Regional-Level Japan Industrial Productivity Database (R-JIP) (in Japanese) <https://www.rieti.go.jp/jp/database/r-jip/>, NITAS (National Integrated Transport Analysis System) (in Japanese) https://www.mlit.go.jp/sogoseisaku/soukou/sogoseisaku_soukou_tk_000055.html, Social and Demographic Statistics System (in Japanese) <https://www.stat.go.jp/data/ssds/index.html>, Social Capital Stock Estimation Data (in Japanese) https://www5.cao.go.jp/keizai2/ioj/result/ioj_data.html, Basic Survey on Employment Structure (in Japanese) <https://www.stat.go.jp/data/shugyou/kako/index.html>, Basic Survey on Wage Structure (in Japanese) <https://www.mhlw.go.jp/toukei/list/chinginkouzou.html>.

Declarations

Conflict of interest The author declares that this manuscript has no conflicts of interest.

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